Cluster Analysis

Hierarchical Clustering

What is Hierarchical Clustering

- This is an agglomerative clustering technique (i.e. bottom-up in nature)
- All the data points, in the beginning, are considered as individual clusters
- The final cluster comprises of all the data points
- Data points are clustered based on similarity/dissimilarity measures
- Usually, Euclidean and cosine distance measures are considered for creating dissimilarity matrix
- Combining clusters depend on linkage function

Linkage Functions

There are several linkage functions available for hierarchical clustering

We will focus on these four commonly used methods

- Single linkage
- Complete linkage
- Group linkage
- Ward method

Single Link Agglomerative Clustering

Use maximum similarity of pairs:

$$sim(c_i, c_j) = \max_{x \in c_i, y \in c_j} sim(x, y)$$

Can result in “straggly” (long and thin) clusters due to chaining effect.

After merging c_i and c_j, the similarity of the resulting cluster to another cluster, c_k, is:

$$sim((c_i \cup c_j), c_k) = \max(sim(c_i, c_k), sim(c_j, c_k))$$

Complete Link Agglomerative Clustering

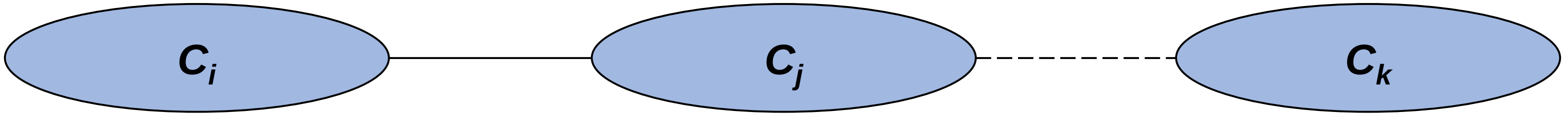
Use minimum similarity of pairs:

$$sim(c_i, c_j) = \min_{x \in c_i, y \in c_j} sim(x, y)$$

Makes “tighter,” spherical clusters that are typically preferable.

After merging c_i and c_j, the similarity of the resulting cluster to another cluster, c_k, is:

$$sim((c_i \cup c_j), c_k) = \min(sim(c_i, c_k), sim(c_j, c_k))$$



Group Average Agglomerative Clustering

Use average similarity across all pairs within the merged cluster to measure the similarity of two clusters

$$sim(c_i, c_j) = \frac{1}{|c_i \cup c_j|(|c_i \cup c_j| - 1)} \sum_{\vec{x} \in (c_i \cup c_j)} \sum_{\vec{y} \in (c_i \cup c_j): \vec{y} \neq \vec{x}} sim(\vec{x}, \vec{y})$$

Compromise between single and complete link

Two options:

- Averaged across all ordered pairs in the merged cluster
- Averaged over all pairs *between* the two original clusters

Some previous work has used one of these options; some the other. No clear difference in efficacy

Ward Method

It says that distance between any two clusters A and B is the increase in sum of square of the merged clusters when compared to individual sum of squares of A and B together

This method tries to club clusters to minimize the increase in sum of square value

Mathematically,

$$
\begin{aligned}
\Delta(A, B) &= \sum_{i \in A \cup B} \|\vec{x}_i - \vec{m}_{A \cup B}\|^2 - \sum_{i \in A} \|\vec{x}_i - \vec{m}_A\|^2 - \sum_{i \in B} \|\vec{x}_i - \vec{m}_B\|^2 \\
&= \frac{n_A n_B}{n_A + n_B} \|\vec{m}_A - \vec{m}_B\|^2
\end{aligned}
$$

Ward’s method is greedy and is applicable for quantitative data only

Example

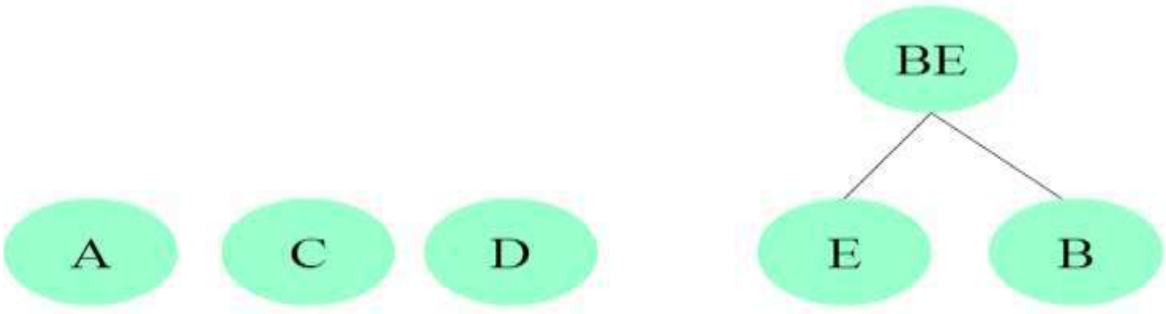
Consider A, B, C, D, E as cases with the following similarities:

	A	B	C	D	E
A	-	2	7	9	4
B	2	-	9	11	14
C	7	9	-	4	8
D	9	11	4	-	2
E	4	14	8	2	-

Highest pair is: E-B = 14

Example

So lets cluster E and B. We now have the structure:



Example

Now we update the case-to-case matrix.

	A	BE	C	D
A	-	2	7	9
BE	2	-	8	2
C	7	8	-	4
D	9	2	4	-

Note: To compute BE -- SC (A, B) = 2

SC (A, E) = 4

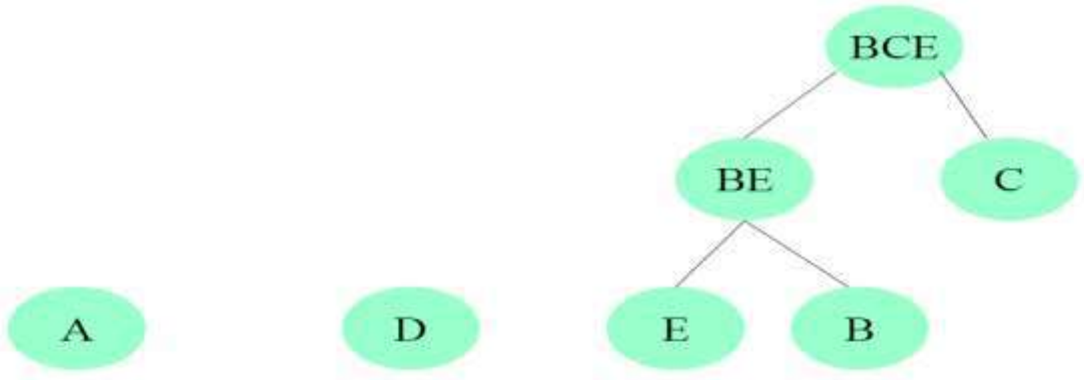
SC(A,BE) = 4 if we are using **single link (take max)**

SC(A,BE) = 2 if we are using **complete linkage (take min)**

SC(A,BE) = 3 if we are using **group average (take average)**

Note: C - BE is now the highest link

Example
so lets cluster BE and C.we now have the structure:



Example

Now we update the case-to-case matrix.

	A	BCE	D
A	-	2	9
BCE	2	-	2
D	9	2	-

To compute SC(A, BCE):

SC (A, BE) = 2

SC (A, C) = 7 so SC(A,BCE) = 2

To Compute: SC(D,BCE)

SC(D, BE) = 2

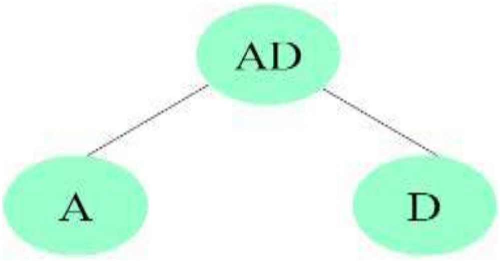
SC(D, C) = 4 so SC(D, BCE) = 2

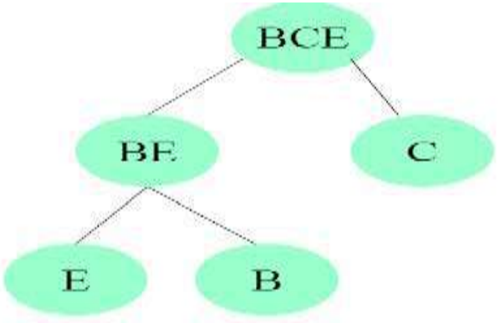
SC(D,A) = 9 which is greater than SC(A,BCE) or SC(D,BCE)

so we now cluster A and D.

Example

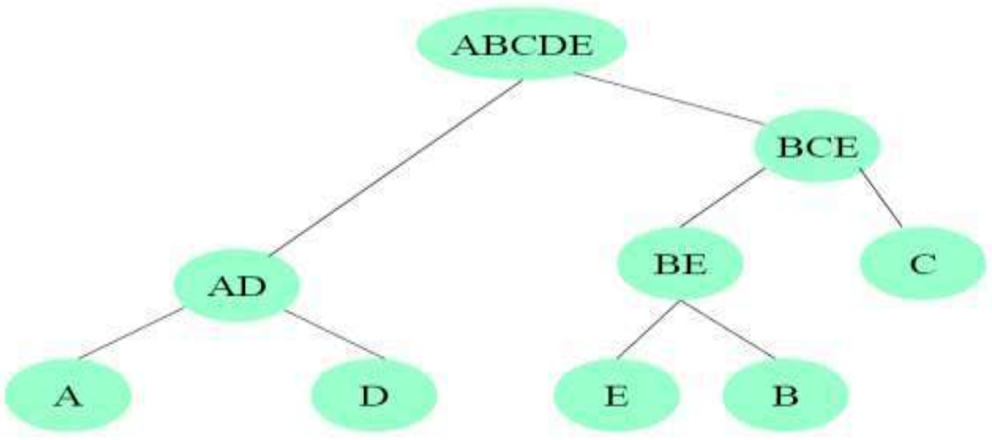
So lets cluster A and D. At this point, there are only two nodes that have not been clustered, AD and BCE. We now cluster them.





Example

Now we have clustered everything



A few more points

To extract proper clusters, careful observation is required while analyzing dendogram

Clusters are formed and get clubbed with other clusters at different heights

Choice of height in cutting the tree (to extract cluster membership), thus requires careful observation

Usually some measures are also extracted while clusters are being formed (such as Cubic Clustering Criteria or CCC) which can help in deciding optimum number of clusters in hierarchical clustering

R codes

Let us use the same dataset what we saw in density based clustering

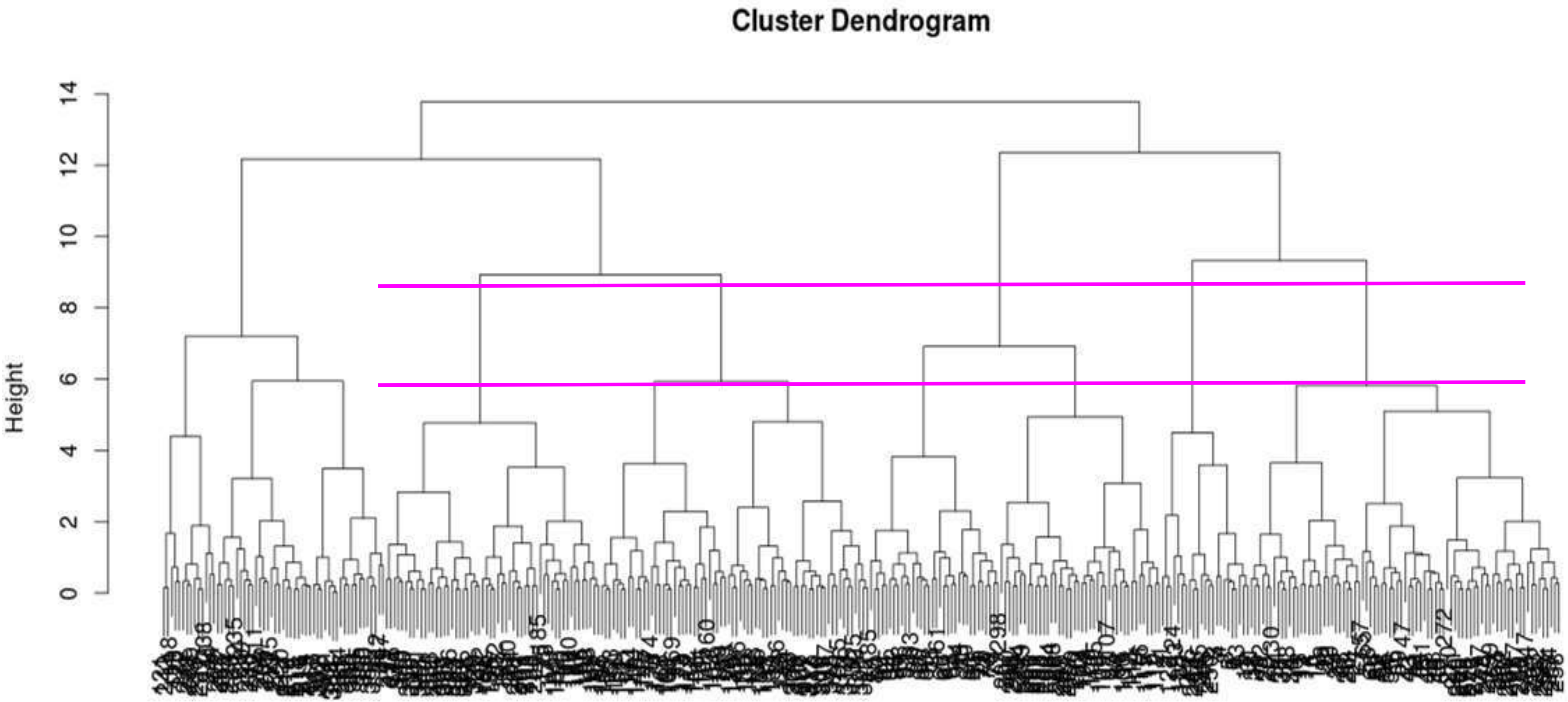
The dataset can be found here also

Hierarchical clustering needs to have dissimilarity matrix as its parameter and a distance matrix can act as a dissimilarity matrix

The following code will do hierarchical clustering of data points (using all default parameters)

```
dist=dist(data)
hc=hclust(dist)
plot(hc)
```

Dendogram

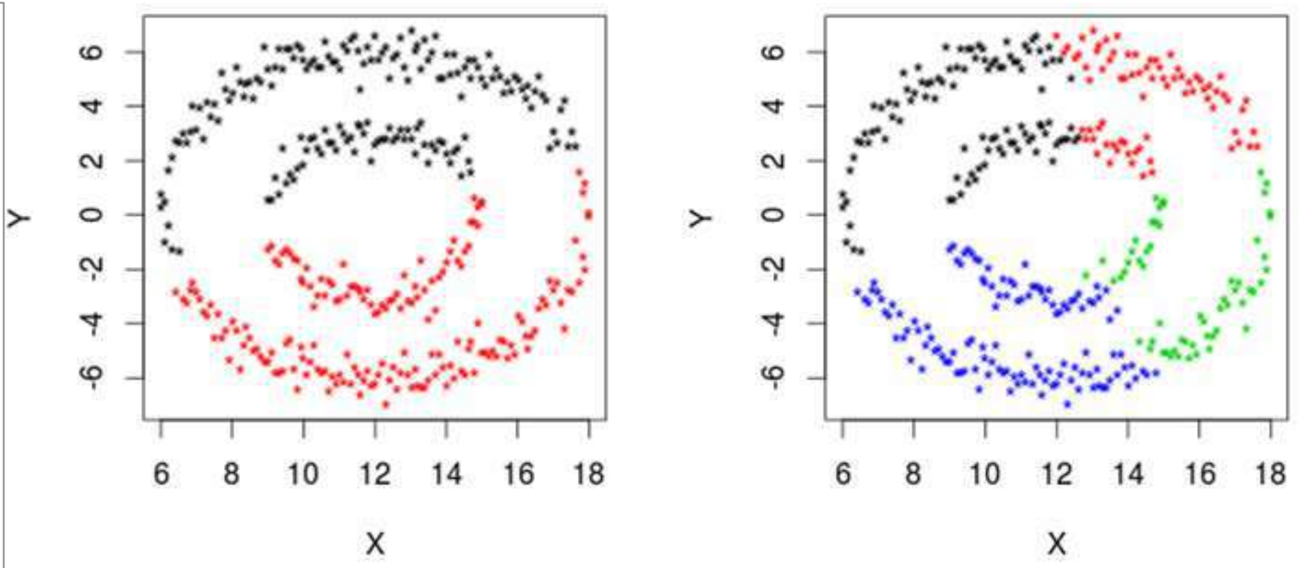


Data Plot

Clustering of data points with 2 cluster and 4 cluster solution

“Complete Linkage” could not bring out the proper clusters

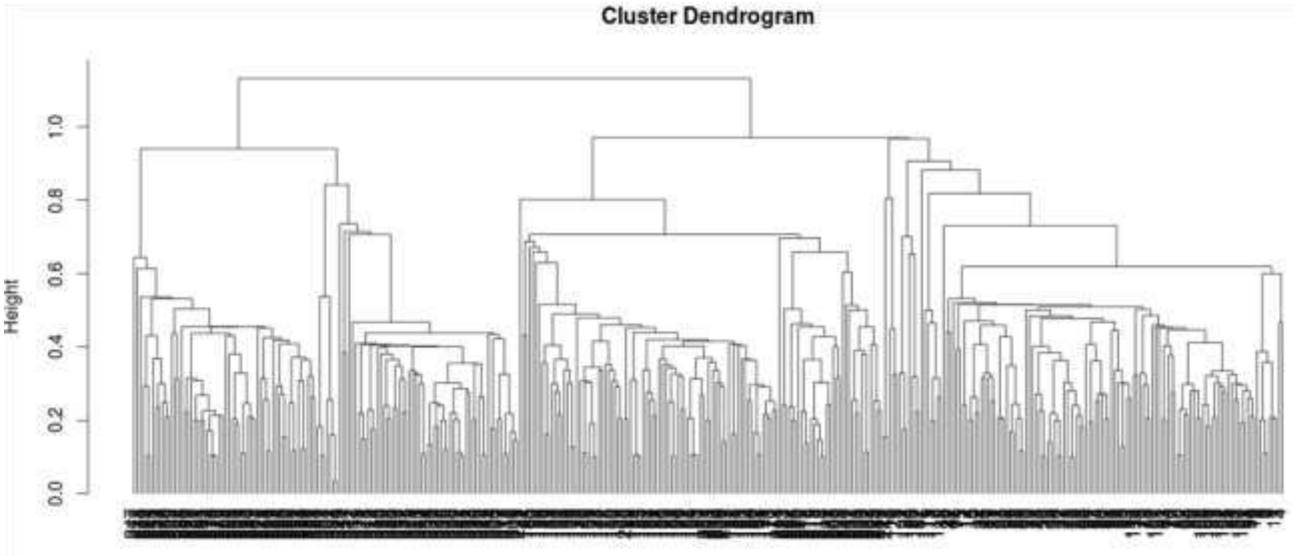
```
dist=dist(data)
hc=hclust(dist,method="complete")
c2=cutree(hc,k=2)
c4=cutree(hc,k=4)
layout(matrix(c(1,2),ncol=2))
plot(data,col=c2,pch="*")
plot(data,col=c4,pch="*")
```



Dendogram (Single Linkage)

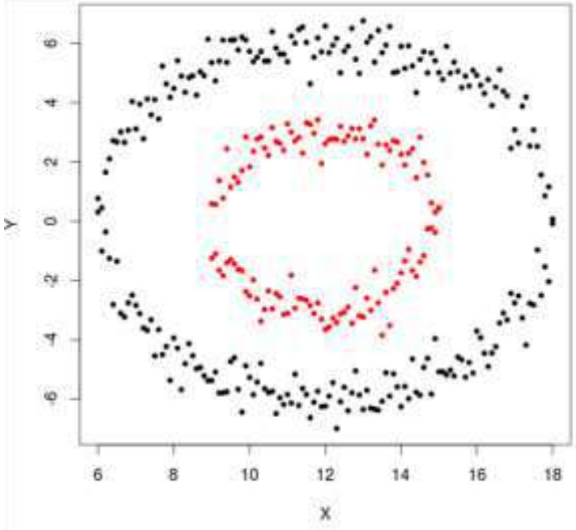
Single Linkage method produces chain like structures and hence the dendogram also looks complicated

However, this method has produced the correct cluster

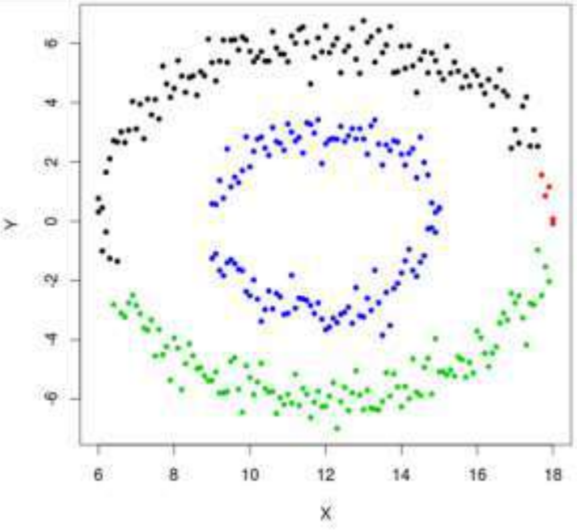


Clustering with Single Linkage

```
dist=dist(data)
hcSL=hclust(dist,method="
single")
k2=cutree(hcSL,k=2)
k4=cutree(hcSL,k=4)
layout(matrix(c(1,2),ncol=2
))
plot(data,col=k2,pch=20)
plot(data,col=k4,pch=20)
```



# of cluster is 2



# of cluster is 4

Advantages and Disadvantages

Advantages of hierarchical clustering are

- Easy to understand
- Often efficient in clustering

Disadvantages of hierarchical clustering are

- Not very much scalable
- Choice of distance measure is far from trivial job
- Choice of algorithm is also not an easy task
- Not applicable for dataset with missing values
- Due to heuristic nature, greedy search may result in unclear cluster hierarchy

Hierarchical Clustering with p-value

Hierarchical clustering can also be associated with uncertainty (i.e. whether the cluster formed is actually a cluster or not)

Normal hierarchical clustering does not produce any such information about the clusters formed

Hence, multiple bootstrapping can be done to asses cluster uncertainty in hierarchical clustering

‘pvclust’ package in R does the same analysis to detect correct clusters

Let us consider a small subset from dataset autompg (first 50 cases only)

Since bootstrapping is required to be done, sample size is kept low (for demo purpose only)

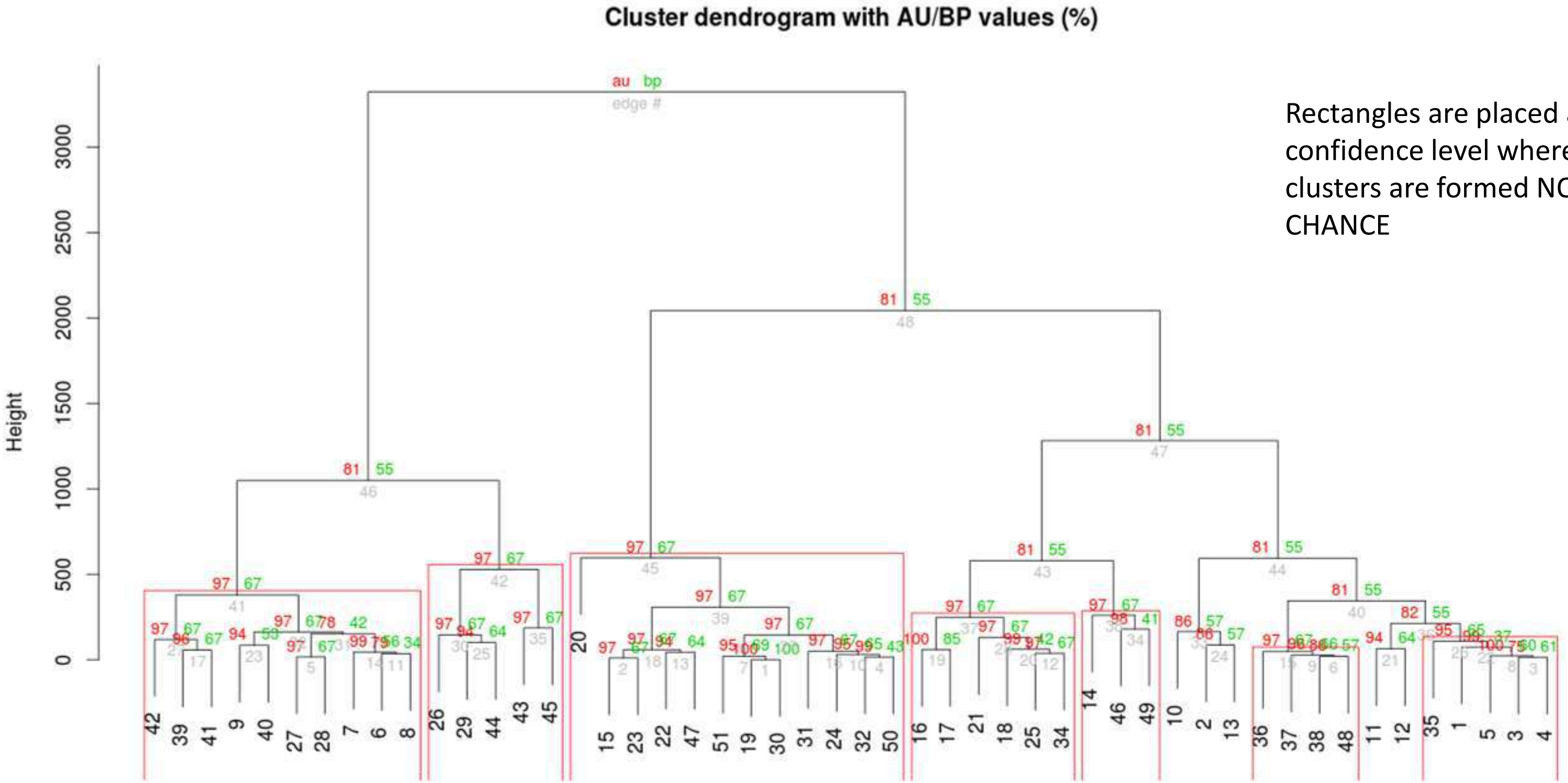
Moreover, only 5 numeric variables are taken into consideration

pvclust, by default, clusters the variables and hence, for clustering cases, transpose of the matrix should be supplied

R Code

```
autompg=read.csv(file.choose())
# Small subset selection
autompg1=na.omit(autompg)[1:50,c(3:6,9)]
library(pvclust)
library(snow) # for parallel computation
cl=makeCluster(10,"MPI") # Creating 10 slaves for parallel computation
# Running pvclust with 1000 bootstrap sampling with different relative sample size
pvclust=parPvclust(cl,t(autompg1),"com","euc",nboot=1000)
plot(pvclust)
pvrect(pvclust,alpha=0.95)
```

Dendogram with p-values



Rectangles are placed at 95% confidence level where the clusters are formed NOT BY CHANCE

Probabilistic Hierarchical Clustering

Probabilistic hierarchical clustering tries to overcome the problems of normal hierarchical clustering by using probabilistic models to measure distance between clusters

It assumes that data are being generated from generative models and each model essentially describe respective clusters

Usually Gaussian distribution of Bernoulli distribution are used in such situations and the objectives lies in estimating the parameters as accurately as possible to generate the same set of data (best fit)

In this sense, it resembles MLE of parameters using Expectation Maximization process

However, for hierarchical clustering, the distances are calculated differently

Let us consider a 1-D dataset (for simplicity)

$$X=\{x_1,x_2,x_3,x_4,x_5,\ldots\ldots x_n\}$$

We can assume that this dataset is generated from a Gaussian distribution with parameters μ and σ

$$N(\mu,\sigma^2)=\frac{1}{\sqrt{2\pi\sigma^2}}exp^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Probability that a point x is generated from this distribution is

$$P(x_i|\mu,\sigma^2)=\frac{1}{\sqrt{2\pi\sigma^2}}exp^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

Consequently, the likelihood function is

$$L(N(\mu,\sigma^2)|X)=P(X|\mu,\sigma^2)=\prod_{i=1}^{n}\frac{1}{\sqrt{2\pi\sigma^2}}exp^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

If the given dataset is partitioned into m different clusters, quality of clustering can be measured by

$$Q(\{C_1, C_2, C_3, \ldots C_m\} = \prod_{i=1}^{m} P(C_i)$$

where P() is the maximum likelihood

Distance between two clusters is given by

$$dist(C_i, C_j) = -log \frac{P(C_i \bigcup C_j)}{P(C_i)P(C_j)}$$

$$log \frac{P(C_i \bigcup C_j)}{P(C_i)P(C_j)}$$

Two clusters keep on merging as long as > 0

Special case

Since, hierarchical clustering needs similarity/dissimilarity measures to cluster objects, hierarchical clustering can be used to cluster data points with mixed data type

However, a different similarity measure is computed using “gower” method

“Gower” similarity measure is not a strict numerical measure and hence, it should not be used in clustering algorithm which assumes data to be numeric in nature (e.g. k-means)

Gower similarity measure

This is a measure which is non metric in nature but can give similarity index between two data points which contain both categorical and numerical data

Gower distance is bound within [0,1]

Gower distance is calculated by taking average of distance measure between ith and jth row considering distances w.r.t. individual variable/attribute

- for categorical attributes: 0 if equal, 1 otherwise
- for ordered attributes: 0 if equal, proportional to rank distance otherwise
- for continuous attributes: proportional to ratio of distance and range of the attribute

Hierarchical clustering with gower dist

In this analysis, we will use the autompg dataset

The first variable is useless and hence deleted

The 2nd, 7th and 8th variable are essentially categorical in nature and hence they are made categorical through type casting

Thus the new dataset becomes a dataset with mixed variables (along with missing data)

To use K-means clustering, we need to remove the factor variables (which may induce significant loss of information)

As an alternative to K-means, we can use either k-medoid or hierarchical clustering with gower distance matrix

If clusters are identified properly, both these methods should give similar output

Since, gower distance is not a strict numeric measure, using ward method is not recommended

Instead, complete linkage is used because it tries to extract spherical clusters

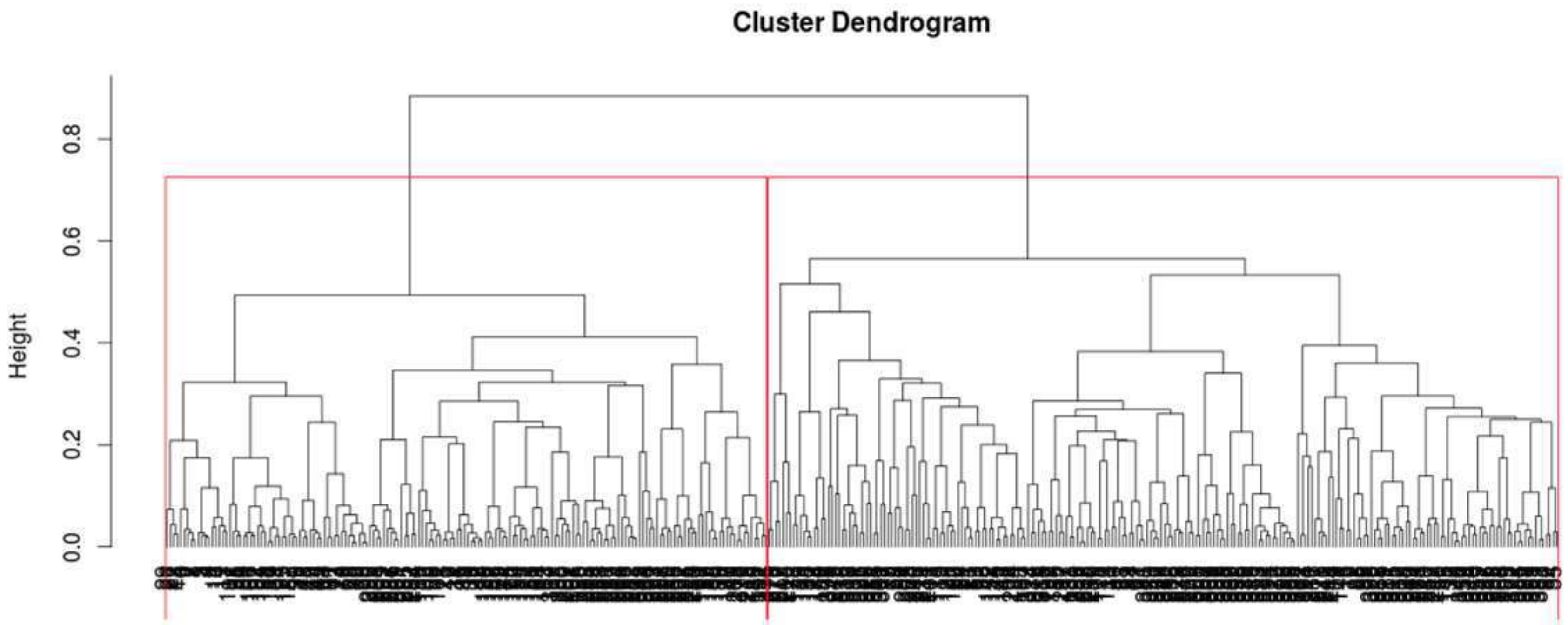
```
# Data Preperation
# Removal of variable X
autompg1=autompg[,-1]

# Converting ‘cylinders’, ‘model’ and
‘origin’ to factors
for(i in c(1,6,7) autompg1[,i]=
as.factor(autompg[,i]
```

```
# Hierarchical clustering with gower distance
library(cluster)
d=daisy(autompg1,metric=’gower’)

# Since gower distance is not a strict numerical
measure, using complete linkage would be a better
option than using ward method
h1=hclust(d,method=’complete’)
plot(h1, hang=-1)

# Two cluster solution
tree=cutree(h1,k=2)
```



Two cluster solution looks more appropriate in this dendogram

Cross check with k-medoid clustering

```r
# K-medoid clustering with number of clusters as 2
clust_pam=pam(autompg1,k=2)

# Creating contingency table
table(clust_pam$clustering,tree)
```

We can see that there is considerable amount of agreement between k-medoid clustering and hierarchical clustering with gower distance

```
> table(clust_pam$clustering,tree)
   tree
      1   2
  1 154  11
  2  18 215
```